

CondorBrain ↔ Gaussian Stack

Alignment Report

(Executive + DevOps teams)

One-Page Executive Summary

CondorBrain serves as the institutional execution, risk, and validation benchmark for all Gaussian-side model development. The system has completed all foundational safety, realism, alignment, and risk layers. Remaining work is limited to deterministic orchestration, final validation, and controlled paper trading.

There is no remaining research risk. Live trading readiness is gated solely by engineering completion and operational verification.

Estimated time to paper-trading readiness once unblocked: *4–5 engineering days*.

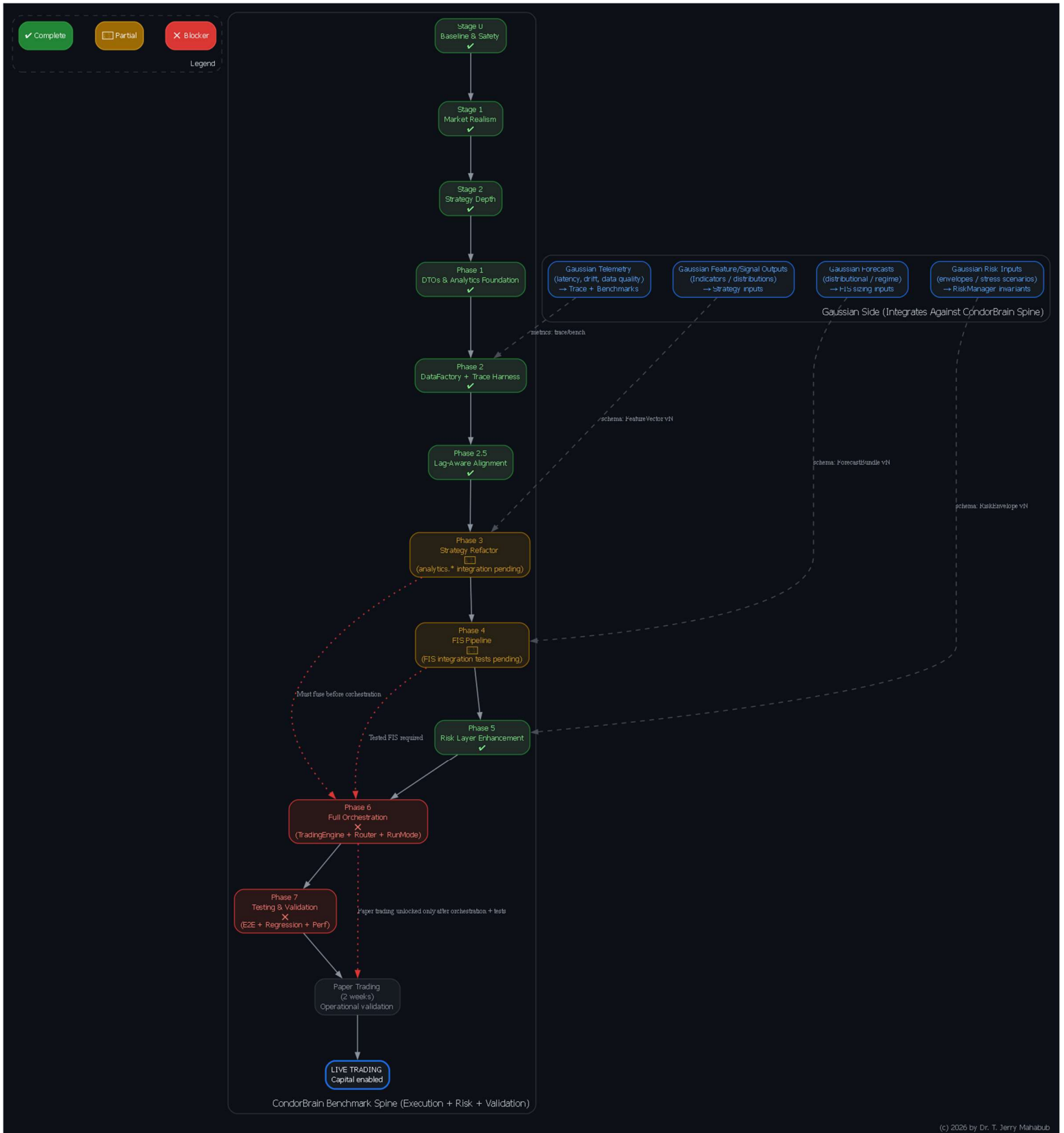
Estimated total time to live trading (including mandatory paper trading): *~3-4 weeks*.

CondorBrain establishes the authoritative execution spine. Gaussian components integrate only through versioned contracts at the Strategy, FIS, and RiskManager boundaries. No Gaussian component bypasses CondorBrain orchestration or validation gates.

High-level path to real-account live trading

Phase	Description	Owner	Duration	Cumulative Time	Status
Phase 3	Strategy + Analytics Integration	CondorBrain	1 day	Day 1	In Progress
Phase 4	FIS Integration Tests	CondorBrain	0.5 day	Day 1.5	In Progress
Phase 6	Full Orchestration + Router	CondorBrain	1–2 days	Day 3–4	Blocked
Phase 7	E2E + Regression Validation	CondorBrain	1 day	Day 4–5	Blocked
Paper Trading	Operational Validation (No Dev)	All	2 weeks	Week 3	Pending
Live Trading	Capital Enabled	All	—	~Week 3+	Gate Dependent

Appendix A — Graphviz DAG



Appendix B — CondorBrain ↔ Gaussian Interface Contract

(Schemas, Invariants, and Operational Guarantees)

Purpose and Scope

This appendix defines the **formal integration contract** between the CondorBrain execution spine and the Gaussian-side analytics, forecasting, and signal-generation components. Its purpose is to ensure that all Gaussian outputs can be safely, deterministically, and auditably consumed by CondorBrain without compromising execution integrity, risk controls, or regulatory defensibility.

CondorBrain is designated as the **benchmark execution system**. Gaussian components are treated as *producers* of signals, forecasts, or auxiliary risk information that must conform to the contracts defined herein.

No live or paper trading is permitted unless all interfaces comply with these contracts and their invariants.

Architectural Principles

- 1. Boundary Enforcement**
All Gaussian outputs enter CondorBrain through explicit, versioned interfaces. There are no implicit couplings or side channels.
 - 2. Fail-Closed by Default**
Any contract violation (schema mismatch, stale data, invalid values) results in trade rejection rather than degraded execution.
 - 3. Time-Correctness First**
Timestamp alignment, lag detection, and decay policies are first-class constraints, not secondary checks.
 - 4. Versioned and Auditable**
Every payload declares its schema version and can be replayed, validated, and regression-tested.
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Integration Surfaces

Feature-to-Strategy Interface

Gaussian may supply numerical features or probabilistic signals used by CondorBrain's strategy gating layer. These inputs must be fully numeric, timestamp-aligned, and explicitly marked for validity when partial data exists.

Forecast-to-Sizing Interface

Gaussian may provide distributional forecasts (means, variances, quantiles, regime probabilities) that influence fuzzy inference sizing decisions. Forecast uncertainty must be explicit and internally consistent.

Risk Envelope Interface

Gaussian may supply external risk constraints or stress scenarios. These are treated as **hard invariants** by CondorBrain's RiskManager and override internal sizing logic when violated.

Telemetry and Diagnostics Interface

Gaussian services must emit telemetry describing latency, data freshness, and operational health. These metrics feed CondorBrain's trace harness and benchmarking framework.

Time Semantics and Alignment

All Gaussian payloads must include:

- Event timestamp (the market time being referenced)
- Production timestamp (when the payload was generated)
- Explicit lag in seconds

CondorBrain enforces symbol-specific lag policies (decay, cutoff, or hybrid). Payloads exceeding allowed lag thresholds are either down-weighted or rejected entirely, depending on configuration.

Invariants (Non-Negotiable)

- No NaN or infinite values may enter CondorBrain unless explicitly masked.
 - Probability distributions must be normalized and internally ordered.
 - Risk limits must be finite and enforceable.
 - Latency and lag must be non-negative and monotonic.
 - Any invariant violation triggers **fail-closed behavior** (no new trades).
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Operational Guarantees

CondorBrain guarantees:

- Deterministic handling of compliant Gaussian inputs
- Traceability from input payload to final order decision
- Enforcement of all risk and alignment constraints

Gaussian guarantees:

- Schema compliance
 - Explicit timestamping and lag reporting
 - Stable versioning and backward compatibility discipline
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Promotion Requirements

Before Gaussian inputs may be used in paper or live trading:

- Contract conformance tests must pass
 - Regression outputs must match golden datasets
 - Latency budgets must be met under load
 - Fail-closed behavior must be demonstrated
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Appendix C — Joint Integration Timeline

(Parallel Workstreams and Promotion Gates)

Objective

This appendix defines a **shared execution timeline** showing how CondorBrain and Gaussian teams can work in parallel while converging safely toward paper trading and live deployment.

CondorBrain acts as the execution and validation backbone; Gaussian work proceeds independently but docks at defined milestones.

Workstreams

- **Workstream A — CondorBrain (Benchmark Spine)**
Execution, risk enforcement, orchestration, validation
- **Workstream B — Gaussian (Analytics & Forecasting)**
Feature generation, probabilistic modeling, regime inference
- **Workstream C — DevOps / Infrastructure**
CI/CD, deployment, observability, broker connectivity

Milestone 1 — Interface Contract Freeze

Objective: Establish stable schemas and invariants.

- CondorBrain publishes the interface contract and validators.
- Gaussian implements payload emitters conforming to the contract.
- DevOps enforces schema checks in CI.

Outcome:

A mutually accepted, versioned contract with sample payloads.

Milestone 2 — Strategy and FIS Docking

Objective: Fuse Gaussian outputs into CondorBrain intelligence layers.

- CondorBrain completes analytics integration and FIS tests.
- Gaussian delivers stable, timestamp-correct features and forecasts.
- DevOps provisions telemetry pipelines and dashboards.

Outcome:

Single-day end-to-end dry runs with full traceability.

Milestone 3 — Orchestration and Routing

Objective: Transition from research flow to execution flow.

- CondorBrain completes TradingEngine, routing, and run modes.
- Gaussian validates operational stability under execution cadence.
- DevOps deploys paper broker adapters and secrets management.

Outcome:

Multi-day backtests using the same pipeline intended for paper trading.

Milestone 4 — Validation Gates

Objective: Prove correctness, stability, and reproducibility.

- CondorBrain executes end-to-end, regression, and performance tests.
- Gaussian passes contract and latency conformance under load.
- DevOps wires CI/CD promotion gates.

Outcome:

A “Paper-Ready” release artifact.

Milestone 5 — Paper Trading

Week 1: Stability Focus

- Latency, disconnect handling, fail-closed verification

Week 2: Economic Reality

- P&L reconciliation, slippage realism, drawdown behavior

Outcome:

Formal Go / No-Go decision for live trading.

Appendix D — Why CondorBrain Is the Benchmark

(Technical Rationale)

Executive Summary

CondorBrain is the benchmark because it already encapsulates the hardest institutional deployment problems: market realism, lag-aware alignment, deterministic risk enforcement, traceability, and auditability. Gaussian innovation is measured under identical execution constraints to ensure observed performance improvements are real, repeatable, and safe to deploy.

Benchmarking Requires More Than Model Accuracy

High-quality models frequently fail in production due to:

- stale or misaligned data,
- unrealistic execution assumptions,
- uncontrolled portfolio risk,
- lack of fail-safe behavior,
- absence of reproducible validation.

CondorBrain explicitly addresses these issues. Gaussian models must therefore demonstrate value **under the same constraints** to be meaningfully benchmarked.

CondorBrain's Institutional Guardrails

CondorBrain already provides:

- Market realism (slippage, commissions, realized volatility)
- Lag-aware alignment with explicit decay and cutoff policies
- Portfolio-level risk aggregation and invariant enforcement
- Deterministic traceability from data to decision
- Structured validation gates before capital exposure

These controls typically represent the largest hidden cost and risk in deployment.

Auditability and Reproducibility

A true benchmark system must support:

- deterministic replay,
- versioned schemas,
- golden regression datasets,
- measurable latency and stability.

CondorBrain's architecture is designed around these properties, enabling credible internal and external review.

Organizational Efficiency

Using CondorBrain as the benchmark:

- Prevents duplicated execution and risk infrastructure
 - Allows Gaussian teams to focus on signal quality
 - Centralizes validation and compliance effort
 - Reduces operational risk during iteration
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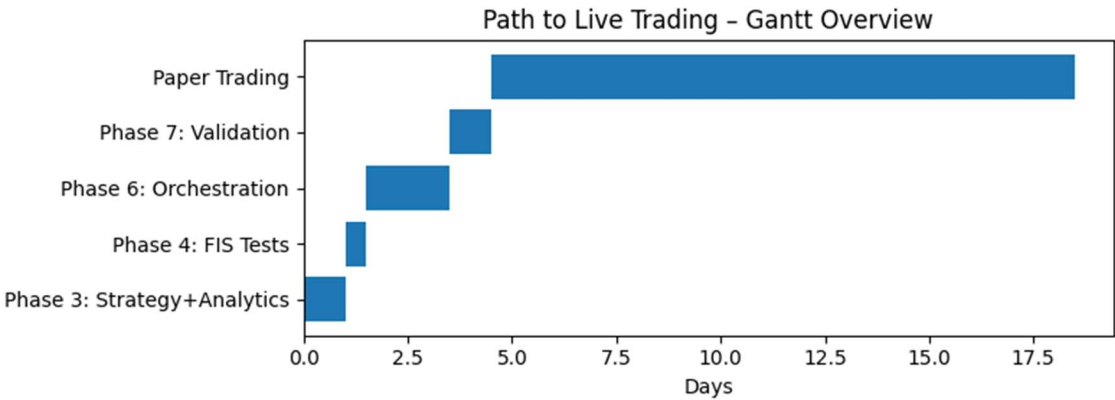
Strategic Conclusion

CondorBrain defines the **deployable standard**:
realistic execution, enforceable risk, time correctness, and testable guarantees.

Gaussian innovation should integrate through CondorBrain's contracts and demonstrate measurable improvement under identical execution constraints. This ensures that any observed performance lift is real, repeatable, and safe to deploy.

Visual Gantt Chart – Path to Live Trading

The chart below provides a visual timeline from engineering completion through paper trading.



Schema Summary Table (Gaussian ↔ CondorBrain)

Schema	Primary Purpose	Key Required Fields	Hard Invariants
FeatureVector	Strategy gating & signals	features, event_ts, produced_ts, lag_sec	No NaN/Inf, lag policy enforced
ForecastBundle	Sizing & regime inference	mean, stdev, quantiles, regime_probs	Quantile ordering, probs sum to 1
RiskEnvelope	Portfolio risk limits	Greek caps, max loss, stress scenarios	Fail-closed on violation

Go / No-Go Sign-Off Page

Live trading authorization requires unanimous approval from the following roles, confirming that all engineering, risk, and validation gates have passed.

Role	Name	Signature / Date
Lead Quant / Architect	Chris Peters	_____
DevOps Lead	John Draper	_____
Risk & Compliance Lead	Jerry Mahabub	_____
Executive Sponsor	Shawn ?	_____

Gaussian Value Measurement Framework

Gaussian-side contributions are evaluated strictly on incremental value added relative to the CondorBrain benchmark, under identical execution, risk, and data constraints.

Primary lift metrics include:

- Risk-adjusted return improvement (Δ Sharpe, Δ Sortino)
- Drawdown reduction under identical exposure
- Stability metrics (lower variance of P&L, fewer tail events)
- Latency and freshness compliance

All lift claims must be demonstrated through paired backtests, paper trading comparison, and regression reproducibility.